Electric Circuits
Year 9

Name _____

Part A

Collect the following electrical components:
power pack, 2 x 12 volt globes, 2 x switches, 1 x rheostat, connecting wires

Task 1 - circuit 1

Connect a circuit in which **one** lamp can be turned on and off by one switch. Note the brightness of the lamp

Draw a circuit diagram to represent the circuit.

Task 2 - circuit 2

Connect a circuit in which **two** lamps can be together switched on and off by **one** switch. Note the brightness of the lamps

Draw a circuit diagram to represent the circuit.

Name _____

Task 3 - circuit 3

Connect a circuit in which **one** switch can turn **two** globes on and off but the globes are as bright as the globe in task 1

Draw a circuit diagram to represent the circuit.

Task 4 - circuit 4

Connect a circuit in which **two** lamps can be switched on and off, each by a **separate** switch. Each lamp can be turned on and off **independently** of the other. Note the brightness of the lamps when they are both on and when only one is on.

Draw a circuit diagram to represent the circuit.

Name _____

Task 5 - circuit 5

Using the circuit from task 4, include in the circuit a **rheostat** so it can be used to alter the brightness of **both** lamps when they are on.

Draw a circuit diagram to represent the circuit.

Questions

1 Which of the circuit(s) is a series circuit ?

2 Which of the circuit(s) is a parallel circuit ?

3 What happens to the brightness of the lamps when more than one are included in a series circuit ?

Part B

In addition to the components you already have, you will need to collect an **ammeter** and a **voltmeter**

Task 6 - circuit 6

Set up **circuit 1** again and this time add to the circuit an **ammeter** which will measure the current flowing in the circuit. Connect the (3A) red terminal of the ammeter to the red terminal on the power pack. The black terminal on the ammeter will be connected to the switch.

What is the reading on the ammeter ?

Draw a circuit diagram to represent the circuit.

Task 7 - circuit 7

Set up **circuit 1** again and this time add to the circuit a **voltmeter** which will measure the potential difference across the lamp. Connect the (3V) red terminal of the voltmeter to one of the lamp terminals and the black terminal of the voltmeter to the other lamp terminal. What is the reading on the voltmeter ?

Draw a circuit diagram to represent the circuit.

Questions

4 What does an ammeter measure ?

5 What does a voltmeter measure ?

6 Is the ammeter connected in series or in parallel with the lamp ?

7 Is the voltmeter connected in series or in parallel with the lamp ?